Suraj Kumar

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GitHub — Portfolio

About Me

Computer Science and AI undergraduate building applied machine learning systems in computer vision and natural language processing. My work spans data preparation, model inference, evaluation, and interfaces for reviewing model outputs.

Education

B.Tech in Computer Science Engineering (Artificial Intelligence) **2024 - 2028**
KIET Group of Institutions, Ghaziabad **CGPA 8.63**
Class XII (CBSE) — Saint Joseph's School NTPC — 2023 **85%**
Class X (ICSE) — Saint Joseph's School Pakartalla — 2021 **87%**

Technical Skills

- **Programming and data** - Python, C++, SQL, pandas, NumPy, Git
- **Computer vision** - YOLOv8, ByteTrack, object detection, multi-object tracking, image preprocessing
- **NLP and retrieval** - MuRIL, DistilBART, MiniLM, TF-IDF, BM25, ChromaDB, CrossEncoder
- **Backend and evaluation** - FastAPI, REST APIs, MongoDB, AWS, ROUGE, F1-score

Experience

Wheeltrix — Data Science & Analytics Intern **Jun - Jul 2026**

- Built TextSummarizer, a web app that ingests TXT, PDF, and DOCX files and generates **extractive summaries with TF-IDF** and **abstractive summaries with DistilBART**.
- Added summary length controls, side-by-side comparison with source text, and **ROUGE evaluation** against a supplied reference summary.

Projects

CVKI — Repository

Goal: Help civic teams prioritize road hazards using submitted photos, GPS coordinates, and local-language reports.

- **Problem:** Citizen reports combine photos, GPS coordinates, and descriptions in local languages for civic teams to review.
- **Approach:** Used YOLOv8 to flag hazards in photos and fine-tuned MuRIL to classify complaints by category, severity, and risk.
- **Workflow:** Sent flagged complaints to a human reviewer to check the reported hazard and its location before routing the case.
- **Technology Used: Python, YOLOv8, MuRIL, FastAPI, MongoDB**

Pedestrian & Vehicle Safety — Repository

Goal: Identify potential conflicts between pedestrians and vehicles in traffic footage near crosswalks and marked risk zones.

- **Approach:** Detected pedestrians and vehicles with YOLOv8 and used ByteTrack to follow their movement across video frames.
- **Dashboard:** Displayed crosswalks and risk zones in FastAPI so reviewers could see how tracked road users moved through them.
- **Technology Used: Python, YOLOv8, ByteTrack, FastAPI**

Urban Spice Events & Catering — Repository — Live Site

Goal: Give a Bangalore catering business a live site where customers can browse menus and services or request a quote.

- **Website:** Built service and menu sections and an event gallery so customers could explore catering options and see past events.
- **Enquiries:** Added a quote request flow and WhatsApp contact option so customers could discuss catering for upcoming events.
- **Technology Used: HTML, CSS, JavaScript**

Core Strengths

Problem decomposition - Split AI projects into testable stages: data preparation, model inference, evaluation, and delivery.

Practical evaluation - Check model outputs against source documents and use ROUGE to measure overlap with reference summaries.

Certifications

- AWS Certified Solutions Architect - Associate — Apr 2026
- AWS Certified AI Practitioner — Mar 2026
- AWS Certified Cloud Practitioner — Dec 2025
- Social Networks — IIT Madras via NPTEL, 12-week course (Jul - Oct 2025)
- AI For Everyone — DeepLearning.AI via Coursera (Mar 2025)

Achievements

- LeetCode — Contest rating **1440** and **200+** problems solved.
- **Flipkart GRiD 8.0** — Advanced to Round 2.
- **Adobe India Hackathon** — Participated in the hackathon.